

- 3 By considering the binomial expansions of $\left(z + \frac{1}{z}\right)^4$ and $\left(z - \frac{1}{z}\right)^4$, where $z = \cos \theta + i \sin \theta$, use de Moivre's theorem to show that

$$\cot^4 \theta = \frac{\cos 4\theta + a \cos 2\theta + b}{\cos 4\theta - a \cos 2\theta + b},$$

where a and b are integers to be determined.

[7]

[illegible]